



COURSE DESCRIPTION FORM

AL-2001 Programming for Artificial Intelligence (PAI) — Laboratory

Institution: FAST School of Computing, National University of Computer and Emerging Sciences, Karachi.

Program: BS Artificial Intelligence — Fall 2026

Course Description

Course Code	AL-2001
Course Title	Programming for Artificial Intelligence
Credit Hours	3 + 1 (Theory + Laboratory)
Prerequisites by Course(s) and Topics	NIL
Grading Policy	Absolute
Policy About Missed Assessment Items in the Course	Retakes for missed assessment items (excluding midterm and final exams) will not be conducted. In the case of a missed midterm or final exam, an exam retake or pre-take application, along with the necessary supporting evidence, must be submitted to the department secretary. The Examination Assessment and Retake Committee will review and decide on all exam re-take and pre-take requests.
Course Plagiarism Policy	Plagiarism in project or midterm / final exam may result in an F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.

Assessment Instruments with Weights

Assessment Item	Weight (%)
Lab Work	20
Midterm Exam	20
Project	10
Final Exam	50

Breakdown of the Lab Work component (20%)

Lab Assessment Item	Weight within Lab Work (%)
In-lab class tasks and demonstration (weekly)	40
Home tasks / lab assignments (weekly submission)	25
Lab Mid-Exam	15



Lab Assessment Item	Weight within Lab Work (%)
Lab Final Exam / project demonstration and viva	20

Course Team and Catalog Description

Head of the Department	Dr. Muhammad Rafi
Course Instructor	Talha Shahid
Lab Instructor	Khadeejah Ashraf
Course Coordinator	—
Current Catalog Description	This course aims to introduce standard programming practices and to help develop programming skills necessary for designing and implementing Artificial Intelligence systems. The course introduces a modern state-of-the-art programming language for Artificial Intelligence, and builds up the necessary programming background for the main courses like Knowledge Representation & Reasoning, Machine Learning, Artificial Neural Networks, and Natural Language Processing. This course will help the students of Artificial Intelligence develop the programming acumen and style. The ultimate aim of this course is to help students in using the programming language to solve problems of interest to them.
Role of the Laboratory	The laboratory runs in parallel with the lecture schedule. Each week a lab manual is issued that restates the concepts covered in the corresponding lecture and turns them into practical work. Every lab session consists of a guided walkthrough, a set of class tasks that must be completed and demonstrated during the session, and a set of home tasks submitted before the announced deadline.

Textbook (or Laboratory Manual for Laboratory Courses)	1. Severance, C. R., 2016. Python for Everybody: Exploring Data Using Python 3. CreateSpace Independent Publishing Platform. 2. Miller, B. N., Ranum, D. L., and Anderson, J., 2019. Python Programming in Context. Jones & Bartlett Learning. 3. McKinney, W., 2012. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython. O'Reilly Media, Inc. 4. Departmental Laboratory Manuals, AL-2001 Programming for Artificial Intelligence (Lab 01 – Lab 14), Department of Artificial Intelligence & Data Science, FAST — NUCES, Karachi, Fall 2026.
Reference Material	1. Joshi, P., 2017. Artificial Intelligence with Python. Packt Publishing Ltd. 2. Janert, P. K., 2010. Data Analysis with Open-Source Tools: A Hands-on Guide for Programmers and Data Scientists. O'Reilly Media, Inc.

A. Course Learning Outcomes (CLOs)

On successful completion of this course, students will know how to:



CLO No.	Course Learning Outcome	Bloom's Taxonomy
01	Comprehend the fundamental constructs of the programming language for data analysis and representation.	Understand
02	Understand and apply the Object-Oriented concepts in the programming language.	Understand
03	Analyze and solve problems using programming, data analysis and standard libraries and/or toolboxes of the programming language.	Analyze

Laboratory alignment. The weekly lab sessions carry the practical assessment of all three CLOs: Labs 01–03 and 05 assess CLO-01, Lab 04 assesses CLO-02, and Labs 07–14 together with the semester project assess CLO-03.

B. Program Learning Outcomes (PLOs)

For each attribute below, a tick indicates that the attribute is covered in this course. A blank cell means the enablement is little or non-existent.

No.	PLO	PLO Detail	Covered
01	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and computing specialization to the solution of complex computing problems.	✓
02	Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems to reach substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✓
03	Design / Develop Solutions	Design solutions for complex computing problems and develop systems, components, and processes that meet specified needs, with appropriate consideration for public health and safety, as well as cultural, societal, and environmental factors.	✓
04	Investigation & Experimentation	Conduct investigations of complex computing problems using research-based knowledge and methodologies.	
05	Modern Tool Usage	Create, select, and apply appropriate techniques, resources, and modern computing tools, including prediction and modeling, to address complex computing problems.	✓
06	Society & Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to the context of complex computing problems.	
07	Environment & Sustainability	Understand and evaluate the sustainability and impact of professional computing work in solving complex computing problems.	
08	Ethics	Apply ethical principles and commit to professional ethics, responsibilities, and norms of computing practice.	✓

No.	PLO	PLO Detail	Covered
09	Individual & Team Work	Function effectively as an individual and as a member or leader in diverse teams and multidisciplinary settings.	✓
10	Communication	Communicate effectively on complex computing activities with the computing community and society at large.	✓
11	Project Management & Finance	Demonstrate knowledge and understanding of management principles and economic decision-making, and apply these to one's own work as an individual or team member.	
12	Life Long Learning	Recognize the need for, and demonstrate the ability to engage in, independent and lifelong learning in the context of ongoing technological changes.	✓

Topics Covered in the Course (15 weeks of instruction)

The right-hand column shows the laboratory session that accompanies each week's lecture topic.

Week	Lecture Topics	Accompanying Laboratory Session
01	Introduction to Python, setting up the development environment (Jupyter Notebooks). Core concepts: variables, expressions, statements, operands, and operators.	Lab 01 — Introduction to Python. Installing Python and Jupyter Notebook, cells, kernel and shortcuts; variables and assignment; expressions, statements, operands and operators; precedence and type conversion; standard data types; decision making; loops.
02	Mastering program flow with conditional statements (if / else) and loops (for, while). Introduction to writing efficient, single-line queries and list comprehensions. In-depth exploration of Python's primary data structures: lists, dictionaries, tuples, and sets — their use cases, methods, and performance characteristics for data handling.	Lab 02 — Control flow and comprehensions. Nested conditions, loop patterns, break / continue, and single-line list and dictionary comprehensions. Lab 03 — Data structures. Lists, tuples, dictionaries and sets; built-in methods; choosing the right structure for a task.
03	Writing custom functions to create reusable and modular code. Arguments, return values, and scope. Introduction to useful built-in functions such as zip and enumerate, and anonymous functions using lambda.	Lab 04 — Functions. Parameters and default arguments, return values, local vs global scope, zip, enumerate, map, filter and lambda.
04	Understanding the principles of OOP: classes, objects, inheritance, encapsulation, and polymorphism.	Lab 05 — Object-oriented programming. Defining classes, constructors, attributes and methods, inheritance and method overriding.
05	Reading from and writing to local files (e.g., CSV, TXT, JSON). Introduction to interacting with web APIs to fetch live data, handle responses, and parse JSON into Python objects.	Lab 06 — File handling and APIs. Reading and writing text, CSV and JSON files; exception handling; consuming a public web API with the requests library.

Week	Lecture Topics	Accompanying Laboratory Session
06	Theory Mid-Exam I	Revision session and a graded in-lab problem-solving task covering Labs 01–06.
07	Introduction to the NumPy library for high-performance numerical operations. Creating and manipulating N-dimensional arrays (vectors, matrices, tensors), array broadcasting, and vectorization.	Lab 07 — NumPy. Array creation, indexing and slicing, reshaping, broadcasting, and vectorized computation against loop-based code.
08	Introduction to the Pandas library for data analysis. Series and DataFrames for data loading, cleaning, filtering, transformation, grouping, and merging.	Lab 08 — Pandas. Loading a dataset, selection and filtering, handling missing values, `groupby`, and merging DataFrames.
09	Principles of effective data visualization. Using Matplotlib to create foundational plots (line, bar, scatter, histogram). Introduction to Seaborn for creating more sophisticated statistical graphics.	Lab Mid-Exam (conducted in the lab slot), followed by Lab 09 — Data visualization with Matplotlib and Seaborn.
10	Machine Learning I	Lab 10 — Machine learning I. The scikit-learn workflow: train / test split, fitting a model, and evaluating predictions.
11	Theory Mid-Exam II	Project proposal submission and progress review in the lab slot.
12	Machine Learning II	Lab 11 — Machine learning II. Model comparison, cross-validation, and reporting results on a small dataset.
13	NLP Option: the programming task of preparing text for AI — text cleaning, normalization, and vectorization techniques (e.g., Bag-of-Words, TF-IDF) using libraries as tools.	Lab 12 — Text processing for AI. Tokenization, cleaning and normalization; building Bag-of-Words and TF-IDF representations.
14	CV Option (image processing): the programming task of handling images — image representation as NumPy arrays, and preprocessing techniques (resizing, colour space conversion, normalization) using libraries such as OpenCV.	Lab 13 — Image processing. Reading images as arrays, resizing, colour space conversion, and normalization with OpenCV.
15	Writing professional, maintainable code (documentation, modularity). A dedicated lecture on Ethical AI (data privacy, bias).	Lab 14 — Professional practice. Code documentation and modularity; final project demonstration and viva.
16	Theory Final Exam	Lab Final Exam.



Programming Assignments Done in the Course	All the assignments would be programming based (Python). Each weekly lab issues class tasks that are completed and demonstrated during the session, and home tasks that are submitted as .py script files under the naming convention RollNo_LabNN_Qn.py (for example 22K-1234_Lab01_Q3.py).
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Class Time Spent (in percentage)

Theory (%)	Problem Analysis (%)	Solution Design (%)	Social and Ethical Issues (%)
25	20	50	5

Oral and Written Communications	Every student is required to submit at least 1 written report of typically 3 pages. Students will also be called for viva / presentation of the project and any assignment where necessary.
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Course Instructor:

Talha Shahid

Signature: _____

Lab Instructor:

Khadeejah Ashraf

Signature: _____

Dated: 19-Aug-2026